



Fig. 2: ARMv7 architecture

ARMv7, which defines the architecture for the first of the Cortex series of cores, defines three architecture profiles as follows:

A-profile (application profile).

ARM Cortex-A cores are intended for application use and consist of ARM Cortex-A5, A7, A8, A9, A12, A15 and A17 MPCore. These are used in devices such as smartphones, tablet PCs, gaming consoles and routers.

R-profile (realtime profile).

ARM Cortex-R cores are intended for robust realtime use and consist of Cortex-R4, R5 and R7. These are used in automotive systems, storage devices, medical instruments, digital TVs, blu-ray players and others.

M-profile (microcontroller profile). ARM Cortex-M cores are intended for microcontroller use and consist of Cortex-M0, M0+, M1, M3 and M4. These are used in automo-

tives, gaming consoles, Bluetooth and ZigBee, touchscreen and power management.

ARMv7 architecture has the following features:

CPU modes. Except in M-profile, the 32-bit ARM architecture specifies several CPU modes, as listed below. At any moment in time, the CPU can be in only one mode, but it can switch modes due to interrupts, or programmatically.

User mode. Usual ARM program execution state

FIQ mode. Mode that is entered whenever the processor accepts an FIQ interrupt

IRQ mode. Mode that is entered whenever the processor accepts an IRQ interrupt

Supervisor mode. Mode that is entered whenever the CPU is reset or an SVC instruction is executed

Abort mode. Mode that is entered whenever a prefetch abort or data abort exception occurs

Undefined mode. Mode that is entered whenever an undefined instruction exception occurs

System mode. Mode that is entered only by executing an instruction

Monitor mode. Mode introduced to support TrustZone extension

Instruction set. The set includes: **Arithmetic instructions.** ARM supports add, subtract and multiply instructions. Integer divide instruc-

tions are only implemented by ARM cores based on their architectures.

Registers. Registers R0 to R7 are unbanked, R13 (stack pointer) and R14 (link register) are banked, and R15 is the program counter. The current program status register (CPSR) holds APSR flags, current processor mode and interrupt disable flags.

Conditional execution. Every ARM instruction has a conditional execution feature called predication, which is implemented with a 4-bit condition code selector called predicate.

Pipelines. ARM7 has a three-stage pipeline; stages being fetch, decode and execute. Cortex-A8 has 13 stages.

Co-processors. Co-processors are used to extend the instruction set. The co-processor space is divided logically into 16 co-processors with numbers from zero to 15.

Debugging. All modern ARM processors include hardware debugging facilities, allowing software debuggers to perform operations such as halting, stepping and breakpointing of code starting from reset. These facilities are built using JTAG support, though some newer cores optionally support ARM's own two-wire SWD protocol. ARMv7 architecture defines basic debugging facilities at an architectural level.

Thumb and Thumb-2. To improve compiled code-density, processors have featured Thumb instruction sets. When in this state, the processor executes the Thumb instruction set, a compact 16-bit encoding for a subset of the ARM instruction set. The space-saving comes from making some of the instruction operands implicit and limiting the number of possibilities, compared to ARM instructions executed in the ARM instruction set state.

Thumb-2 extends the limited 16-bit instruction set of Thumb with additional 32-bit instructions to give the instruction set more breadth, thus producing a variable-length instruction set. A stated aim for Thumb-2 was to achieve code density similar to Thumb with performance similar

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